# RevIN Experiment 2: Normalized Backpropagation

## 1 Theoretical overview and goal

With instance statistics $(\mu_x, \sigma_x)$, data-space MSE is $\|\hat{y} - y\|^2$, whereas normalized backpropagation minimizes

$$\mathrm{nMSE}(\hat{y}, y \mid x) = H^{-1} \sum_{h=1}^{H} \left( \frac{\hat{y}_h - y_h}{\sigma_x + \varepsilon} \right)^2 .$$

The latter gives comparable gradient weight to low- and high-scale windows. Predictions are still denormalized and evaluated with ordinary data-space MSE, so only optimization weighting changes.

**Experimental goal.** Test whether scale-balanced gradients improve generalization independently of the forward normalization layer, and whether RevIN's affine parameters alter that effect.

## 2 Implementation details

Two paired contrasts are run:

| Forward transform | Data-space loss | Normalized loss |
|---|---|---|
| Non-affine instance | `instance_mse` | `instance_nmse` |
| Affine RevIN | `revin_mse` | `revin_nmse` |

Architecture, seed, batch composition, learning rate, and epoch budget are held fixed within every pair. The project supports MSE, MAE, nMSE, nMAE, and relative MSE, but this paper comparison focuses on MSE versus nMSE.

**Loss implementation:** `src/utils/pipeline.py`
**Experiment orchestration:** `src/scripts/experiment.py`

The normal sweep covers six datasets, seven settings, DLinear/PatchTST, and five seeds, with validation every 1,000 steps and horizon-strided evaluation. Final tables always use test MSE in data space, show sample standard deviations, and state the row multiplier. Macro summaries give each dataset–setting pair equal weight and include a test-selected standard/instance oracle only as an optimistic reference.

## 3 Interpretation

An nMSE-trained model can win on data-space MSE because balanced gradients learn a mapping that generalizes across heterogeneous scales. Training curves from MSE and nMSE have different units and should not be compared by absolute height; selection should use the common validation metric.